

REBUILT KICKOFF

Scoring Analysis Workshop

In 2026, 3468 hosted Kickoff at West Monroe High School for the 1st time with 6 other FRC teams traveling from Shreveport, LA, to Vicksburg, MS, to Nachitoches, LA.

After the Rebuilt live stream, our teams took a lunch break and returned to a deep dive into the Rebuilt Game Manual. Our team began to brainstorm robot design after analyzing the scoring components and ranking points.

Initial Robot Action Priority List:

- Climb in the Auto
- Climb to Level 3
- Fast Single Shooter
- L2 Swerve Drivetrain
- Hooper Goal: (30 - 40 balls)
- Ground Pick-up: In-built bumper
- Travel over the bump and under the trench

Ice Storm Fern Delays...

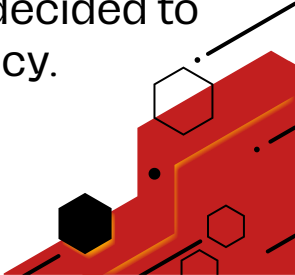
After school was cancelled for an entire week early during the build timeframe, we had to adjust our priorities.

Robot Action Reprioritized List:

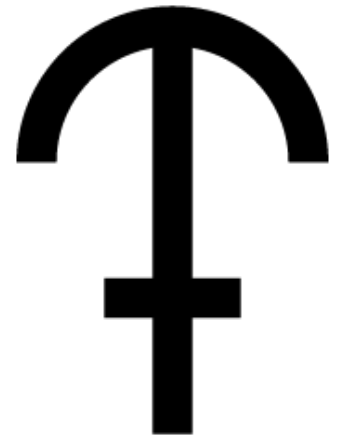
- Fast Single Shooter
- L2 Swerve Drivetrain
- Extendable Hopper (20-30 balls)
- Ground Pick-up: In-built bumper
- Travel over the bump and under the trench

Arkansas Regional - KRONOS 2.0

Our robot did not work the way we intended. The drive team had to switch to a defense strategy, and upon returning from AR, the team decided to build a modified EVERYBOT shooter/intake to increase efficiency.



KRONOS 1.0 VS. 2.0



KRONOS

3468 HP



GREEK ROBOT. FRAME: 25 X 30", WEIGHT: 100 LBS.

Ability AETHER

KRONO'S LIGHT WEIGHT FRAME ALLOWS THE ROBOT TO GLIDE THROUGH TRENCHES AND SOAR OVER BUMPS



FUEL BARAGE

GIVES THE ROBOT THE ABILITY TO DISPURSE FUEL IN NOT ONLY THE HUB BUT ACROSS THE FIELD

OFFENSE 🚀 X3 DEFENSE 🚀 X4 EST. 🚀 FEB. 21ST



@frcmagnatech3468



@magnatech_3468



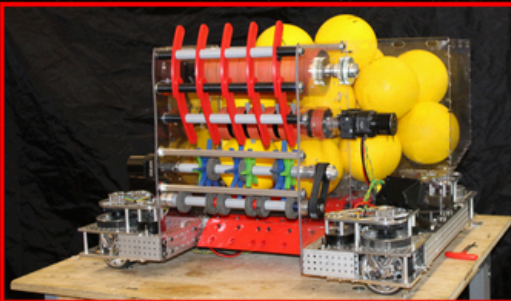
@frc3468.MAGNatech



@MAGNatech 3468 Robotics

KRONOS 2.0

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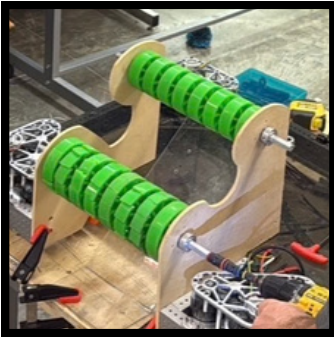


@MAGNatech 3468 Robotics

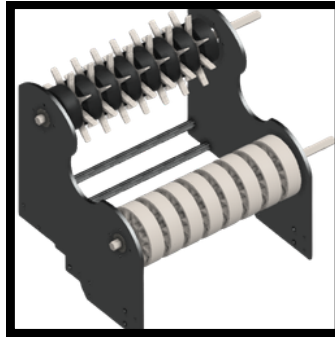


KRONOS 1.0: INTAKE/SHOOTER

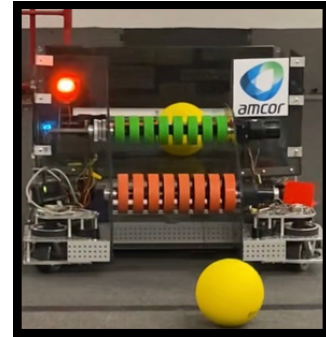
1ST DESIGN



CAD



FINAL DESIGN



Description: Our intake system is a dual subsystem that works as an intake and a shooter. The lower wheels grip the fuel and push it back into the hopper.

When ready to score, the upper wheels then shoot the fuel back while pinching it between both sets of wheels to control movement.

Improvements: The intake design was easy to modify, allowing the team to improve performance and address issues.

The system was updated to include two rows: the first with green 35A compliant wheels and the second with a shooting mechanism using a combination of 35A compliant stars and wheels.

Strength: This design combines the intake and shooting mechanism into one subsystem, saving space on the robot frame and allowing quick transitions.

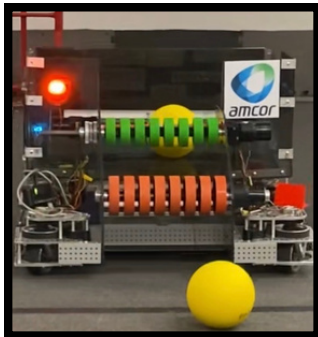
Weakness: ~~We could definitely shoot balls quicker and intake multiple balls at a time. A stretch goal of ours would be to build and program a turret during the offseason.~~

KRONOS 2.0: INTAKE/SHOOTER

1ST DESIGN

CAD

FINAL DESIGN



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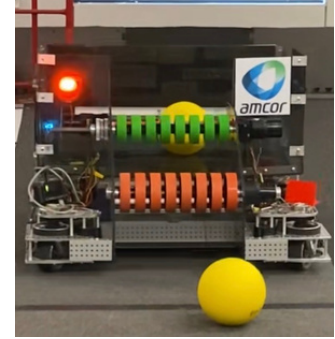
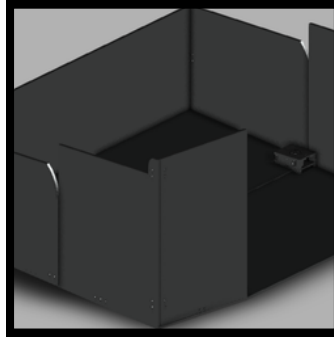
Improvements: Brass flywheels have a higher inertia for more efficient shooting accuracy.

Strength: This design combines the intake and shooting mechanism into one subsystem, saving space on the robot frame and allowing quick transitions.

Weakness: We could definitely shoot balls quicker and intake multiple balls at a time. A stretch goal of ours would be to build and program a turret during the offseason.

SUBSYSTEM # 2:

KRONOS 1.0 EXTENDABLE HOPPER



Description: We chose an extendable hopper that can expand past the robot's frame perimeter once the match begins. At the start of the game, the hopper remains within the frame perimeter to follow the rules. When collecting fuel, the hopper expands to increase storage capacity. The system holds about 8 fuel at the start of the match and can expand to hold approximately 35 fuel, allowing the robot to collect and store more balls during gameplay.

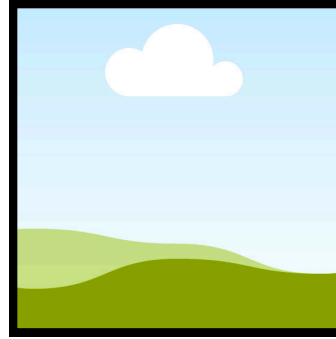
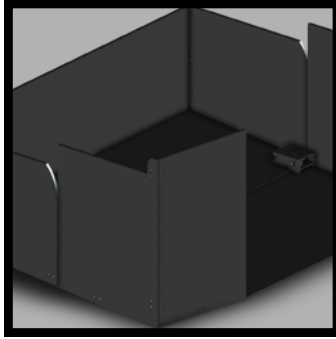
Improvements: A future offseason goal is to increase the hopper capacity to about 50 fuel while maintaining reliable ball movement. We also plan to design a more integrated hopper and intake subsystem to improve efficiency and control fuel flow. Additionally, making the hopper less wide could create enough space on the robot to include a climbing mechanism while still maintaining a large-capacity fuel storage system.

Strength: The extendable hopper improves scoring efficiency by allowing the robot to store a larger amount of fuel before returning to score. By expanding during collection, the drive team can spend less time traveling between zones. The design also helps keep fuel moving through the system, especially with the added agitator and hopper movement, which improves ball flow and supports more consistent shooting performance.

Weakness: During initial prototype testing, we discovered that two balls could become stuck at the same time, causing jams in the hopper. This required adjustments to the design, including adding an agitator and using hopper movement to keep the fuel in motion. The extendable design also required us to prioritize space for the hopper, which meant we chose not to include a climbing mechanism on the robot.

SUBSYSTEM # 2:

KRONOS 2.0 FIXED HOPPER



Description: We chose an extendable hopper that care balls during gameplay.

Improvements: A future offseason goal is to increase the

Strength: The extendable hopper improves scoring efficien.

Weakness: During initial prototype testing, we discovered that two balls

AUTONOMOUS CODE

Changes from last code:

The main change we made to autos was that we synced the bot up more with pathplanner so we could have faster and more accurate movements.

What can it do?

It can go anywhere on the field in the most efficient way with no driver manipulation. It can shoot and pick up balls automatically.

Strengths:

Scoring in auto can give our team an advantage because we can start with a higher score and score extra points.

Opportunities for Improvement:

The biggest opportunity for improvement would be to incorporate cameras into our auto code so that the robot's positioning is more accurate

TELE-OP CODE

Coding Process:

In terms of coding styles we had a much larger team to work on small pieces and compile them together in one folder.

Key Features: Multiple rookies worked on subsystems and commands with assistance from veterans. Our rookies are able to understand the components of our code inside subsystems and commands.

Strengths:

The ability to use a limelight to line up with the hub for distance and direction accuracy to the hub using AprilTags.

Opportunities for Improvement:

Improve the code to have more accuracy with shooting fuel and make more conventional build choices for teleop.

DRIVER STATION CONTROLS

Changes from last code:

We improved the robot's code by adding the ability to align with scoring elements using more AprilTag detection. This allows the robot to recognize field markers and automatically adjust its position for more accurate alignment. The update helps the robot consistently position itself when approaching scoring locations, improving reliability and reducing manual correction needed from the driver during matches.

Key Features:

Our subsystem controller includes thirty-six interchangeable buttons that can be re-mapped each season to perform different robot functions. Thirty-two buttons are located on the top in a square pattern with the corners removed, and additional buttons are placed on the side for quick access. The large number of buttons allows the operator to control many subsystems and actions efficiently. This design also provides a unique and customizable way of controlling the robot.

Strengths:

Previously, drivers had difficulty translating their perspective on the field to the robot's orientation, which made precise movement challenging. The joystick-based control system helps solve this issue by making movement inputs more intuitive and responsive. With directional controls and orientation alignment built into the code, drivers can more easily maneuver the robot, improving accuracy, consistency, and overall driving performance during gameplay.

Improvement: One improvement is to add a second Thrustmaster Joystick to the driver station to improve control of the swerve drivetrain. With two joysticks, the driver could control left and right movement with one joystick and front and back movement with the other, similar to the control style of an Xbox Controller.



A "NEW ERA" CUSTOM ROBOT CART

The "New Era" project involved replacing a rusted, non-functional industrial hydraulic lift with a custom-built, modular transport workstation. This transition was driven by the need for mechanical reliability, integrated electronics, and a professional team aesthetic.

The Case for Improvement

- **Mechanical Obsolescence:** The original hydraulic lift had failed, preventing necessary height adjustments for robot maintenance.
- **Logistical Inefficiency:** Lack of storage forced team members to hand-carry driver stations and tools through crowded venues.
- **Aesthetic Disconnect:** The "rusty red" appearance undermined the team's goal of professionalizing their pit presence and branding.

Feature	Engineering Benefit	Operational Impact
Modular Rail System	High adjustability for varying robot frame perimeters.	Longevity: Cart remains viable across multiple game seasons.
Integrated Docking	Secure mount for the Driver Station laptop and controllers.	Risk Mitigation: Prevents damage to critical control electronics.
Foldable Chassis	Collapsible sections for transport and pit storage.	Efficiency: Optimizes limited 10'x10' pit space.
Onboard Power Hub	Integrated charging ports and 3-prong outlets.	Readiness: Ensures 100% uptime for scouting and drive tablets.
Greek-Themed Imagery	Custom vinyl wraps and synced Alliance-color LEDs.	Branding: Creates a cohesive, high-end "Imagery" package.



INTEGRATED BATTERY MANAGEMENT & DIAGNOSTIC STATION

Following a critical failure during the Charged Up! season—where the team suffered multiple brownouts due to unidentified defective batteries—we engineered a custom-built Battery Management Station. This station is a centralized hub for high-capacity charging and computerized health diagnostics, ensuring that every battery entering the robot is flight-certified and reliable.

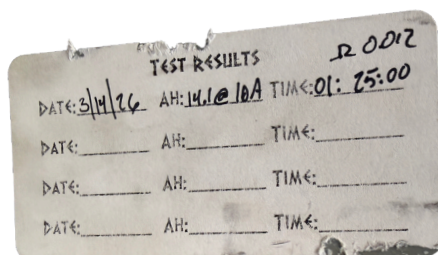
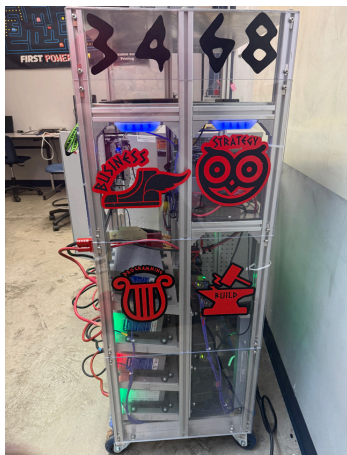
The Case for Improvement (The "Hard Lesson Learned")

During a previous competition, the team struggled with recurring brownouts on the field. It was only after borrowing equipment from Team 16 (Bomb Squad) that we discovered our batteries were internally defective.

Lack of Data: Previously, the team had no way to distinguish between "strong" match-ready batteries and "weak" practice batteries, leading to unpredictable robot performance.

Charge Inefficiency: Standard chargers were disorganized and difficult to manage in a fast-paced pit environment, risking uncharged batteries being placed in the robot.

Feature	Engineering Function	Team Impact
8-Port Charging Bay	Rapidly charges a full fleet of batteries simultaneously.	Readiness: Guarantees a fresh, 100% charged battery is available for every back-to-back match.
Computerized Diagnostic Port	Integrates a battery tester with a top-mounted computer system.	Reliability: Allows students to record and analyze discharge data to "retire" old or defective batteries.
Integrated Station Link	Provides a dedicated power outlet to charge the Universal Drive Station.	Continuity: Ensures the driver's laptop never loses power between matches or during tethering.
Greek-Themed Branding	Finished with custom Greek-themed vinyl wraps.	Imagery: Transforms a piece of utility equipment into a professional part of the team's visual brand.



DRIVER STATION CONTROLS

